PROJECT PRESENTATION

sEMG

MUSCLE FATIGUE



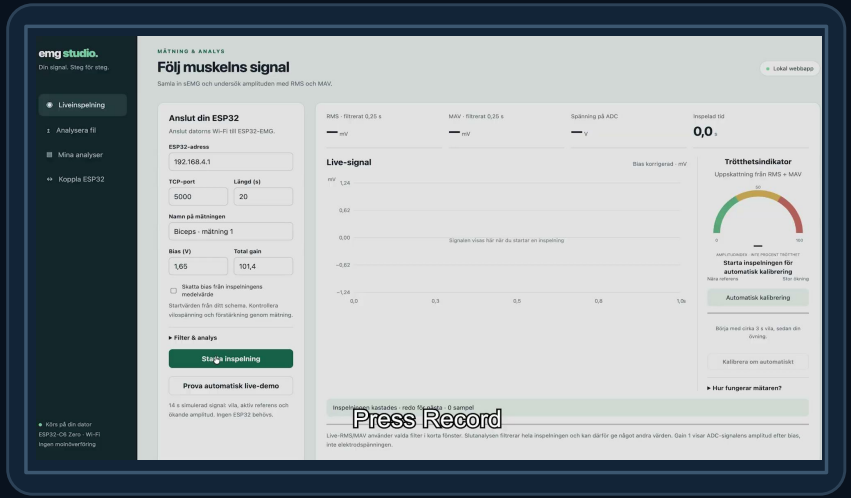
AZIZ, MELVIN, ASSEM, ADOM / 26-10-21

Project Introduction

PROJECT OVERVIEW

- Electrodes measures voltage in your muscle
- Circuit amplifies & filters signal
- Microcontroller transfer data
- Computer analyzes & calculates data
- Application shows results

PROJECT VIDEO



INTENDED USER

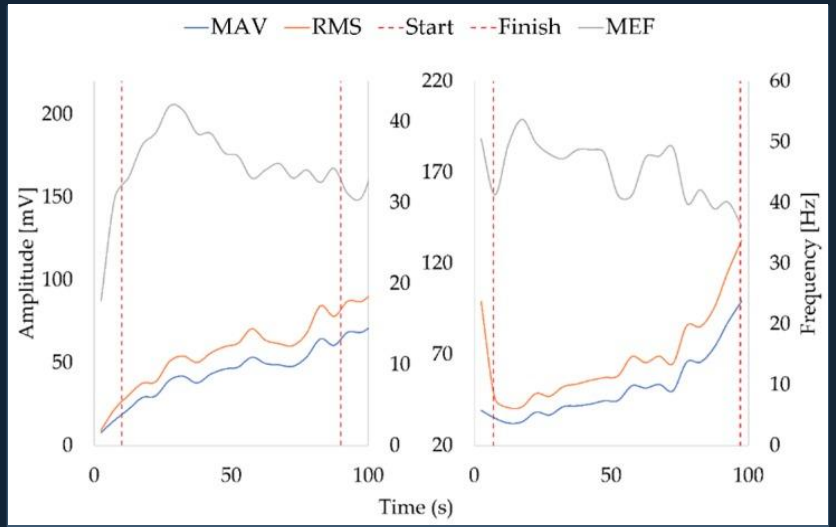
Problem Statement

- Signal affected by noise
- Small amplitude
- Electrode placement

Previous Research

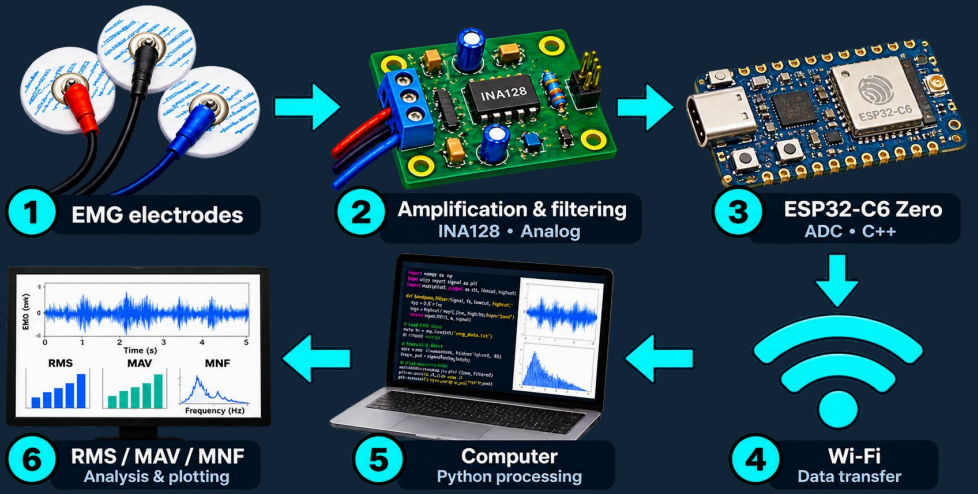
- Muscle fatigue → RMS | MAV | MNF
- Primary method

PROBLEM



VISUAL REFERENCE

Technical Approach



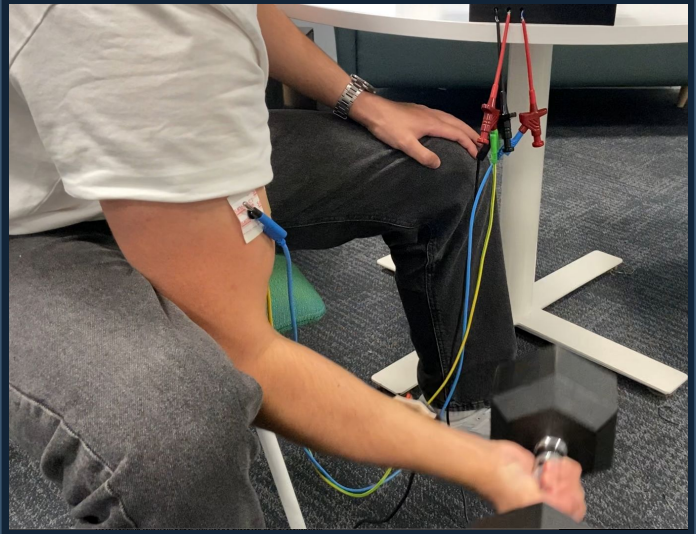
Amplify small signal

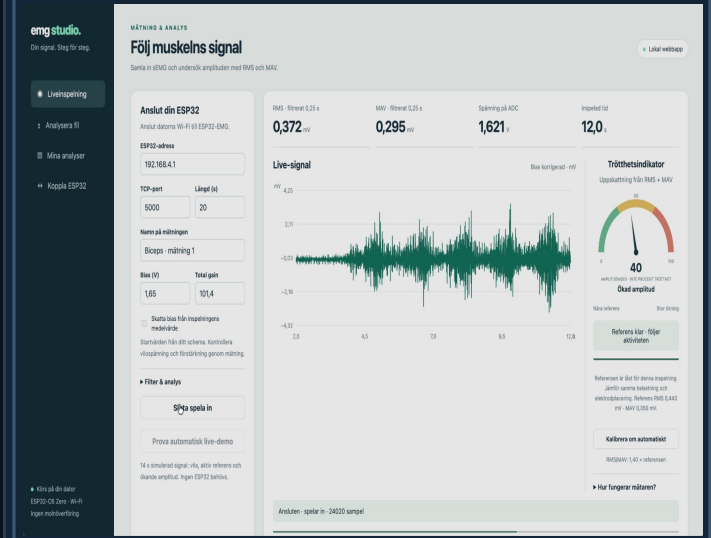
Computer processing

Reliable Electrodes

SYSTEM DESIGN / CIRCUIT / BLOCK DIAGRAM

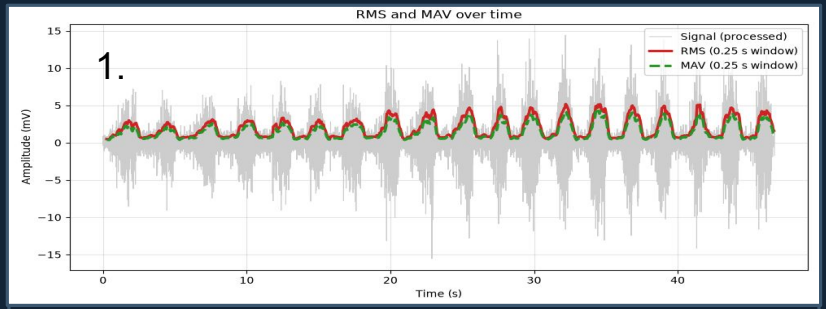
User Interaction & Experience

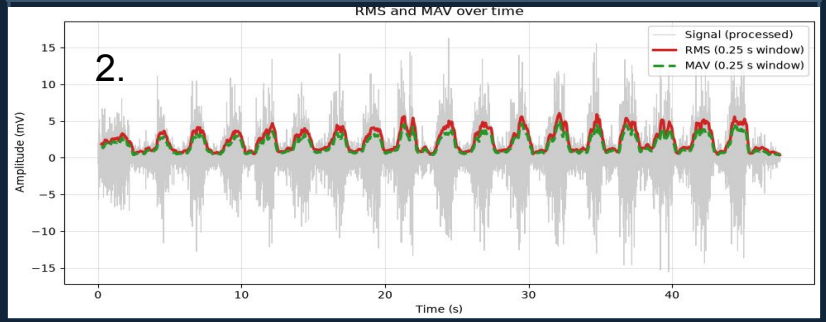


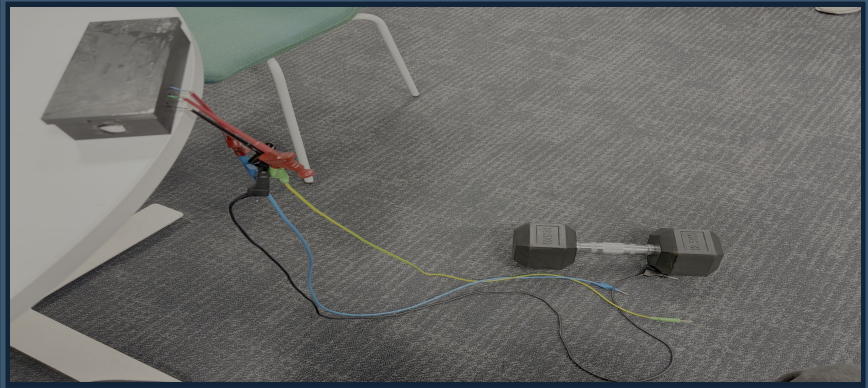


1. Place the electrodes
2. Press record
3. Exercise
4. Observe Results

Results & Prototype







- Measuring electrode: Yellow & Blue cable
- Reference electrode: Black cable

Project Methodology

WHAT WENT WELL

- Data Transfer
- Measurement
- Python program

WHAT WE WOULD CHANGE

- Rechargeable batterier
- On and of switch
- Compact & lighter
- Better electrodes



THANK YOU!